

7 (short) A two-bit counter implemented using D-type bistables is shown in Fig. X.

(a) Draw the state diagram for the two-bit counter. [5]

(b) What is the code implemented by this counter and why is this advantageous in certain applications? [2]

(c) How would you reconfigure the counter to reverse the sequence determined above? Draw the actual circuit implementation. [3]

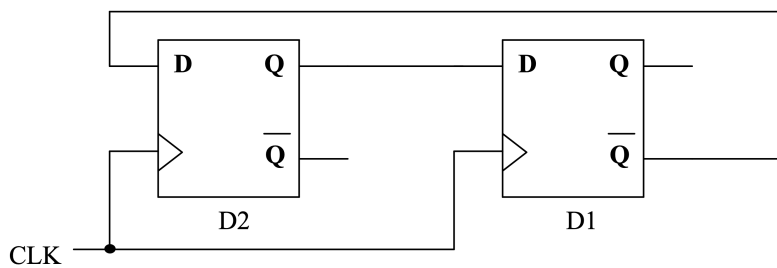


Fig. 5